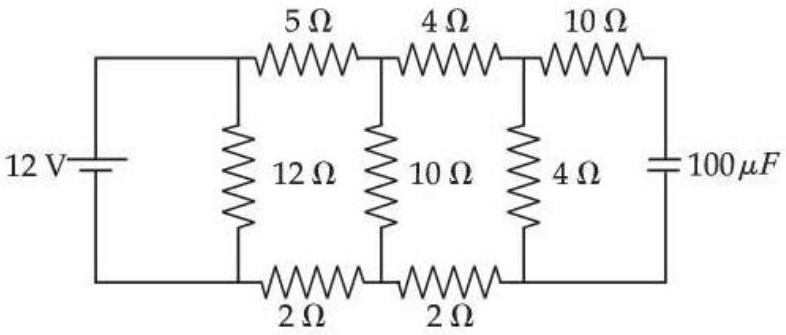
The stored charge in the capacitor in steady state of the following circuit is _____ μC.



Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 50 Question Id : 691121575 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Two masses of 3.4 kg and 2.5 kg are accelerated from an initial speed of 5 m/s and 12 m/s, respectively. The distances traversed by the masses in the 5th second are 104 m and 129 m, respectively. The ratio of their momenta after 10 s is $\frac{x}{8}$. The value of x is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69112147
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112147
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 691121576 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I		**List - II**
	Mass of substance		**Number of atoms**
A.	1.8 mg water	I.	$2\times10^{-4}\times N_A$
B.	9.8 mg sulphuric acid	II.	$1.5\times10^{-4}\times N_A$
C.	1.8 mg carbon	III.	$3\times10^{-4}\times N_A$
D.	5.85 mg salt (NaCl)	IV.	$7\times10^{-4}\times N_A$

Choose the **correct** answer from the options given below :

Options :

6911211956. A-IV, B-III, C-I, D-II

6911211957. A-III, B-II, C-IV, D-I

6911211958. A-III, B-IV, C-II, D-I

6911211959. A-III, B-IV, C-I, D-II

Question Number : 52 Question Id : 691121577 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Given : Molar mass of C, H, O, Cl are 12, 1, 16 and 35.5 g mol^{-1}, respectively

Statement I : In 30% (w/w) solution of methanol in CCl_4(at T K), the mole fraction of CCl_4 is equal to 0.33.

Statement II : Mixture of methanol and CCl_4 shows positive deviation from Raoult's law.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211960. Both **Statement I** and **Statement II** are true

6911211961. Both **Statement I** and **Statement II** are false

6911211962. **Statement I** is true but **Statement II** is false

6911211963. **Statement I** is false but **Statement II** is true

Question Number : 53 Question Id : 691121578 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Bromine trifluoride autoionizes to form $BrF_2^{\oplus}$ and $BrF_4^{\ominus}$. The shapes of the cation and anion are respectively _____, and _____.

Options :

6911211964. bent, square planar

6911211965. linear, square planar

6911211966. bent, see-saw

6911211967. linear, tetrahedral

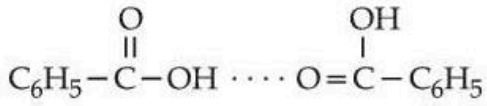
Question Number : 54 Question Id : 691121579 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following statements are **not correct** ?

A. For water, magnitude of K_b is more than the magnitude of K_f.

B. The elevation in boiling point of water when a non-volatile solute is added to it is larger in magnitude than its depression in freezing point.

C. Osmotic pressure measurement is preferred over any other colligative property to determine molar mass of proteins and polymers.

D. The dimerised form of benzoic acid in benzene is 

Choose the **correct** answer from the options given below :

Options :

6911211968. A and B only

6911211969. A and D only

6911211970. A, B and D only

6911211971. A, C and D only

Question Number : 55 Question Id : 691121580 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following reactions in which all the reactants and products are present in gaseous state

$2\,xy \rightleftharpoons x_2 + y_2 \quad K_1 = 2.5 \times 10^5$

$xy + \frac{1}{2} z_2 \rightleftharpoons xyz \quad K_2 = 5 \times 10^{-3}$

The value of K_3 for the equilibrium $\frac{1}{2} x_2 + \frac{1}{2} y_2 + \frac{1}{2} z_2 \rightleftharpoons xyz$ is :

Options :

6911211972. 2.5×10^{-3}

6911211973. 2.5×10^{3}

6911211974. 1.0×10^{-5}

6911211975. 5×10^{-3}

Question Number : 56 Question Id : 691121581 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given at 298 K : $E^{\ominus}_{Fe^{2+}/Fe} = X$ Volt

$E^{\ominus}_{Fe^{3+}/Fe} = Y$ Volt

The $E^{\ominus}_{Fe^{3+}/Fe^{2+}}$ in Volt at 298 K is given by :

Options :

6911211976. $2X - 3Y$

6911211977. $3Y - 2X$

6911211978. $3Y + 2X$

6911211979. $Y + X$

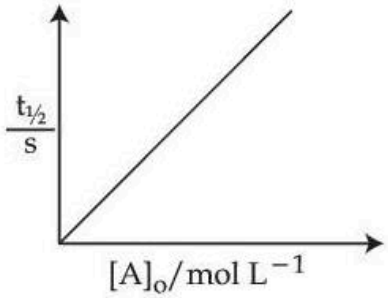
Question Number : 57 Question Id : 691121582 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

$R = 8.314$ J K^{-1} mol^{-1} and 1 cal = 4.2 J

Statement I : When Ea = 12.6 kcal/mol, the room temperature rate constant is doubled by a 10 °C increase in temperature (298 K to 308 K)

Statement II : For a first order reactions A → B,



Here $[A]_o$ is the initial concentration of A and $t_{1/2}$ is half life of reaction.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211980. Both **Statement I** and **Statement II** are true

6911211981. Both **Statement I** and **Statement II** are false

6911211982. **Statement I** is true but **Statement II** is false

6911211983. **Statement I** is false but **Statement II** is true

Question Number : 58 Question Id : 691121583 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match **List - I** with **List - II**.

	List - I Electronic configuration of neutral atom (where n = 2)		**List - II** 1st Ionization Energy (kJ mol^{-1})
A.	ns^2	I.	2080
B.	ns^2np^1	II.	899
C.	ns^2np^3	III.	800
D.	ns^2np^6	IV.	1402

Choose the **correct** answer from the options given below :

Options :

6911211984. A-II, B-III, C-IV, D-I

6911211985. A-IV, B-III, C-II, D-I

6911211986. A-III, B-II, C-IV, D-I

6911211987. A-III, B-II, C-I, D-IV

Question Number : 59 Question Id : 691121584 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Find the **correct** statements related to group 15 hydrides.

A. Reducing nature increases from NH_3 to BiH_3

B. Tendency to donate lone pair of electrons decreases from NH_3 to BiH_3

C. The stability of hydrides decreases from NH_3 to BiH_3

D. HEH bond angle decreases from NH_3 to SbH_3 (E = Elements of group 15)

Choose the **correct** answer from the options given below :

Options :

6911211988. A and B only

6911211989. B and C only

6911211990. A, B, C and D

6911211991. A, C and D Only

Question Number : 60 Question Id : 691121585 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : The number of pairs among $[Ti^{4+}, V^{2+}]$, $[V^{2+}, Mn^{2+}]$, $[Mn^{2+}, Fe^{3+}]$ and $[V^{2+}, Cr^{2+}]$ in which both ions are coloured is 3.

Statement II : The number of pairs among $[La^{3+}, Yb^{2+}]$, $[Lu^{3+}, Ce^{4+}]$ and $[Ac^{3+}, Lr^{3+}]$ ions in which both are diamagnetic is 3.

In the light of the above statements, choose the **correct** from the options given below :

Options :

6911211992. Both **Statement I** and **Statement II** are correct

6911211993. Both **Statement I** and **Statement II** are incorrect

6911211994. **Statement I** is correct but **Statement II** is incorrect

6911211995. **Statement I** is incorrect but **Statement II** is correct

Question Number : 61 Question Id : 691121586 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements for catalytic properties of transition metals.

Statement I : First row transition metals which act as catalyst utilise their 3d electrons only for formation of bonds between reactant molecules and atoms on the surface of catalyst.

Statement II : There is increase in the concentration of reactants on the surface of catalyst which strengthens the bonds in reacting molecules.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911211996. Both **Statement I** and **Statement II** are correct

6911211997. Both **Statement I** and **Statement II** are incorrect

6911211998. **Statement I** is correct but **Statement II** is incorrect

6911211999. **Statement I** is incorrect but **Statement II** is correct

Question Number : 62 Question Id : 691121587 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Vapours of the liquid with higher boiling point condense before vapours of the liquid with lower boiling points in fractional distillation.

Statement II : The vapours rising up in the fractionating column become richer in high boiling component of the mixture.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911212000. Both **Statement I** and **Statement II** are true

6911212001. Both **Statement I** and **Statement II** are false

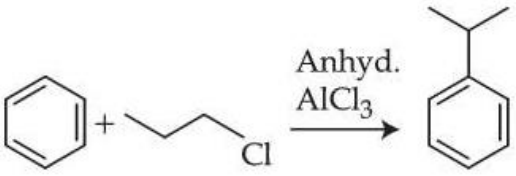
6911212002. **Statement I** is true but **Statement II** is false

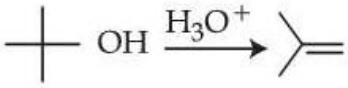
6911212003. **Statement I** is false but **Statement II** is true

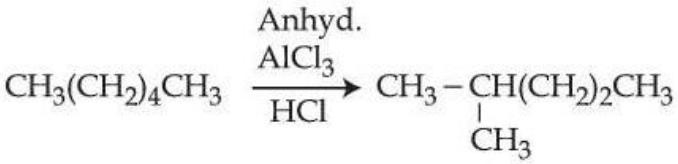
Question Number : 63 Question Id : 691121588 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

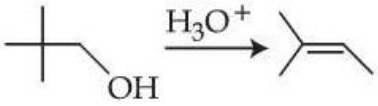
The major product of which of the following reaction is not obtained by rearrangement reaction ?

Options :

6911212004. 

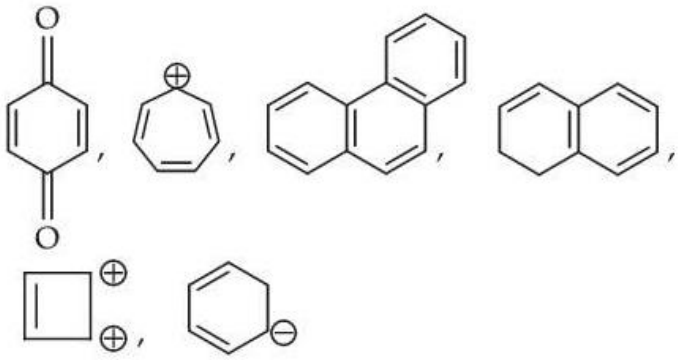
6911212005. 

6911212006. 

6911212007. 

Question Number : 64 Question Id : 691121589 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The total number of aromatic compounds/species from the following is



Options :

6911212008. 6

6911212009. 4

6911212010. 3

6911212011. 5

Question Number : 65 Question Id : 691121590 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

n-Butane on monochlorination under photochemical condition gives an optically active compound "P". "P" on further chlorination gives dichloro compounds.

The number of dichloro compounds obtained (ignore stereoisomers) is :

Options :

6911212012. 3

6911212013. 4

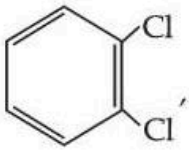
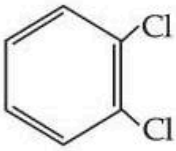
6911212014. 5

6911212015. 6

Question Number : 66 Question Id : 691121591 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Due to increase in van der Waals forces, the order of boiling points is $CH_3CH_2CH_2I > CH_3CH_2I > CH_3I$.

Statement II : As  is more symmetric, its melting point is higher than  , however its boiling point is lower than 

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

6911212016. Both **Statement I** and **Statement II** are true

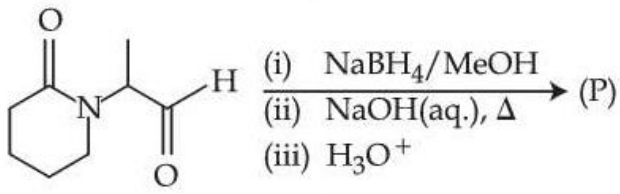
6911212017. Both **Statement I** and **Statement II** are false

6911212018. **Statement I** is true but **Statement II** is false

6911212019. **Statement I** is false but **Statement II** is true

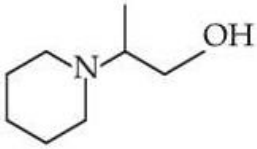
Question Number : 67 Question Id : 691121592 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

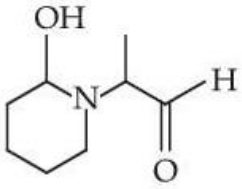
Consider the following reaction.

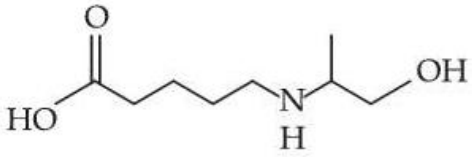


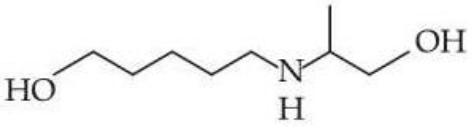
The major product (P) formed is :

Options :

6911212020. 

6911212021. 

6911212022. 

6911212023. 

Question Number : 68 Question Id : 691121593 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which statements are **True** ?

A. In Hoffmann bromamide degradation, 4 moles of NaOH and 2 moles of Br_2 are consumed per mole of an amide

B. Hoffmann bromamide reaction is not given by alkyl amides.

C. Primary amines can be synthesized by Hoffmann bromamide degradation.

D. Secondary amide on reaction with Br_2 and NaOH will give secondary amine.

E. The by-products of Hoffmann degradation are Na_2CO_3, NaBr and H_2O.

Choose the **correct** answer from the options given below :

Options :

6911212024. A, C and E only

6911212025. B, C and D only

6911212026. C and E only

6911212027. C, D and E only

Question Number : 69 Question Id : 691121594 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The **incorrect** statement from the following with respect to carbohydrates is :

Options :

6911212028. All monosaccharides are reducing sugars.

6911212029. The monosaccharide units obtained from hydrolysis of oligosaccharides are always the same.

6911212030. Starch and cellulose are typical examples of polysaccharides, which are very high molecular weight compounds of more than ten monosaccharide units.

6911212031. Open chain and cyclic structures co-exist at equilibrium that are responsible for certain properties as in the case of D − (+) − glucose.

Question Number : 70 Question Id : 691121595 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following amino acid will give violet coloured complex with neutral ferric chloride solution ?

Options :

6911212032. Threonine

6911212033. Serine

6911212034. Tyrosine

6911212035. Cysteine

Chemistry Section B

Section Id :	69112148
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0

Sub-Section Number : 1
Sub-Section Id : 69112148
Question Shuffling Allowed : Yes
Is Section Default? : No

Question Number : 71 Question Id : 691121596 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Number of paramagnetic complexes among the following is _____.
$[MnBr_4]^{2-}$, $[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$, $[CoF_6]^{3-}$, $[Fe(CN)_6]^{4-}$, $[Mn(CN)_6]^{3-}$, $[Ti(CN)_6]^{3-}$, $[Cu(H_2O)_6]^{2+}$, $[Co(C_2O_4)_3]^{3-}$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 691121597 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

'x' is the product which is obtained from benzene by reacting it with carbon monoxide and hydrogen chloride in the presence of cuprous chloride. 'y' is the major product obtained from the benzene by reacting it with ethanoyl chloride in the presence of anhydrous $AlCl_3$. Product (major) obtained by heating x and y in the presence of alkali is z. Total number of π (pi) electrons in z is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 691121598 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider two radiations of wavelengths

1. $\lambda_1 = 2000$ Å
2. $\lambda_2 = 6000$ Å

The ratio of the energies of these two radiations $\left(\frac{E_1}{E_2}\right)$ is _____ (Nearest integer).

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 691121599 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the reaction

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

The magnitude of enthalpy change for the reaction in kJ mol^{-1} is _____. (Nearest integer)

Given : $\Delta_f H^{\ominus}(H_2S) = -20.1\ kJ\ mol^{-1}$

$\Delta_f H^{\ominus}(H_2O) = -286.0\ kJ\ mol^{-1}$

$\Delta_f H^{\ominus}(SO_2) = -297.0\ kJ\ mol^{-1}$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 691121600 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Solid carbon, CaO and $CaCO_3$ are mixed and allowed to attain equilibrium at T K.

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ $\quad Kp_1 = 0.08$ atm

$C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ $\quad Kp_2 = 2$ atm

The partial pressure of CO is _____ $\times 10^{-1}$ atm

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1